

CMOR 504

Project Report

Project Report

*Non-Isomorphic Local Complement Branch and Bound Scheme and its
Applications to Graph Theory Fundamentals*

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Abstract

Recent work in quantum computing focuses on understanding qubits in the lense of graph theory. One such operation includes the local complement. Despite its simple definiton, it can be become a redundant task if the graph is presumed to be dense and have local complements that are isomorphic to previous graphs. We're focused on constructing a recursive algorithm that finds all non-isomorphic graphs from the initial-state graph. Given the final listing of all the non-isomorphic graphs, we'll proceed to quantify the various properties of each graph including power domination, maximum stable set, cliques, matchings, and perfect matchings.

Introduction

Graphs act as a source of understanding of[1]

Preliminary Material

Text on the preliminary material goes here

Methodology and Results

Text on the methodology and results goes here

Conclusions

Text on the conclusions material goes here

Limitations, Discussions, and Future Work

Text on the limitations, discussions, and future work goes here

Bibliography

- [1] Jeremy C. Adcock, Sam Morley-Short, Axel Dahlberg, and Joshua W. Silverstone. Mapping graph state orbits under local complementation. *Quantum*, 4:305, August 2020.